

Permission to Dance: An End-to-End Dance Enhancement System from Dance Capture to Analysis

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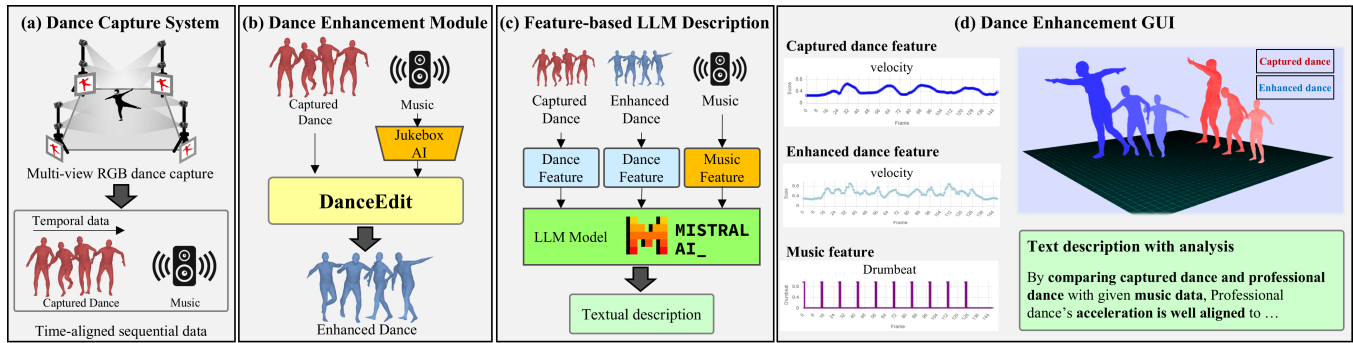


Figure 1: Overall pipeline of end-to-end dance capture, enhancement and analysis system

Abstract

In this demonstration, we present Permission to Dance, an end-to-end dance enhancement system designed to capture, enhance, and analyze user's dance performance. Our system consists of a dance capture module, a dance enhancement module, and a dance feedback module. Using the system, users can acquire their dance data in an enhanced version, followed by textual feedback on how to achieve better dance performance. The demonstration video is available at <https://youtu.be/lFw7Xic48KU>.

CCS Concepts

• Human-centered computing → Gestural input; • Computing methodologies → Computer vision.

Keywords

Dance enhancement, Large Language Model, Real-time Capture

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1 Introduction

Dancing is a natural and intuitive human response to music, enabling individuals to express their **personal intentions**. However, intention alone typically results in amateur dance performance, which falls short of professional standards. Rhythmic alignment between music and choreography, as well as technical proficiency in dance, are required for professional dance. However, achieving this is extremely difficult without a high level of expertise. To provide support to the amateur dancer, we propose Permission to Dance, an end-to-end system designed to capture amateur dance performances and enhance them to a professional level. Our system first captures the user's dance, and then generates an enhanced choreography using DanceEdit, which enhances the dance to professional quality while preserving the user's intention. After that, we compare the features of both the captured and enhanced dances with the music features, which quantifies the music and choreography alignment. Finally, the LLM generates textual feedback based on the feature analysis, offering users clear guidance for further improvement. We implement the system as a Graphical User Interface, allowing users to conveniently use the system. Through this interface, users can view both their original dance and its professional-level enhancement, as well as receive detailed textual analysis of the improvements.

2 System Design

2.1 Dance Capture System

In our system, we adopt the multi-view dance capture framework of Kim et al. [4]. First, we perform multi-view 3D pose estimation by independently applying CLIFF [8] to each RGB camera and extracting SMPL [9] parameters. We leverage the NVIDIA TensorRT [11] engine, which improves the overall speed and accuracy of the dance capture for real-time capture. Finally, to address the self-occlusion problem, we fused SMPL parameters to represent the user's dance with high accuracy.

2.2 Dance Enhancement Module

DanceEdit draws inspiration from the principles of SDEdit [10], a guided image editing framework. SDEdit enables semantic editing by injecting noise into user guidance, such as stroke painting, and then applying a pretrained diffusion model [1]. This process does not require explicit training on paired data or a guidance dataset. This allows high-quality image editing from simple structural guidance while preserving the original structure. Analogously, DanceEdit adapts this approach to the domain of dance, operating on SMPL dance sequence $\mathbf{x}_0 \in \mathbb{R}^{T \times D}$, where T is the number of frames and D is the pose dimension, instead of images. Here, we set the guidance as an amateur dance, which often lacks the technique and rhythmic alignment of a professional dance. We leverage EDGE [13] as a pretrained diffusion model, trained on a professional dance dataset AIST++ [7]. DanceEdit enhances amateur dance without the need for an amateur dance dataset or additional model training.

DanceEdit achieves enhancement through the following procedure. First, it injects a fixed amount of Gaussian noise into the SMPL dance sequence $\mathbf{x}_0$ to produce a noised version $\mathbf{x}_t$ that preserves the intention of dance. EDGE then initiates the denoising process from $\mathbf{x}_t$ to $\hat{\mathbf{x}}_0$, generating a professional dance while maintaining the original intention. We empirically set the denoising timestep t to 400, balancing structural consistency and proficiency in the results. In previous work, amateur dance dataset is typically created by manipulating the temporal and spatial properties of professional dances. However, such synthetic data cannot fully capture the genuine intention of amateur dances, as this process often disrupts the original intention and expressiveness inherent in amateur dance. In contrast, by leveraging the properties of diffusion models, DanceEdit preserves the intention of the original dance throughout the enhancement process, making it more effective for real-world dance enhancement.

2.3 Feature-based LLM Description

As discussed in Section 1, a dance is perceived as professional when the music is properly aligned with the choreography. To assess improvements in the enhanced dance, we extract and compare features from the enhanced dance, the original dance, and the music. For music, we focus on rhythmic and structural elements [3, 6] such as BPM, segment, waveform, and drumbeat. For the dance, we draw inspiration from Laban Movement Analysis, characterizing the dance using features such as velocity, acceleration, motion space, trajectory speed, kinetic energy, and isolation [5, 12]. The music features are compared with both the enhanced dance features and

Table 1: Comparison of dance enhancement models

Method	FID ↓	Beat Align. ↑
Zhou et al. [14]	24.461	0.21
Ours	21.942	0.25

Table 2: Subjective test with users

Method	Intention ↑	Proficiency ↑
EDGE [13]	2.6	4.3
Zhou et al. [14]	3.8	4.1
Ours	4.2	4.3
Pro dancer	4.7	4.8

the original dance features. The comparison result is then used as input to the Mistral [2] LLM, which generates textual descriptions that provide insights into the dance enhancement. These analyses are seamlessly integrated into our system GUI, enabling users to intuitively compare the original and enhanced dances, feature graphs, and LLM-generated explanations.

3 Experiments

We conduct experiments to evaluate how well the enhanced dance matches the intention of amateur dance and the quality of professional dance. Here, we use EDGE [13] a pretrained model for DanceEdit. To the best of our knowledge, the only previous work on dance enhancement is that of Zhou et al. [14], so we use it as a quantitative comparison. For a realistic evaluation, we collect test data from the dance capture system performed by amateur dancers. As shown in Table 1, we benchmark our method against Zhou et al. [14] using a test set of amateur dance sequences. Five participants performed amateur dances to ten different AIST songs, and each performance was recorded for 10 seconds. The data is then processed by each model to generate enhanced dance samples. For evaluation, we use FID score with AIST++ dataset and Beat Alignment as metrics. The results show that DanceEdit achieves better performance than the baseline model. Given that subjective perception is a critical aspect of dance quality, we further conduct a user study. The results are shown in Table 2, based on two standards for professional dances. The first question assesses how well the enhanced dance reflects the intention of the original dance, while the second question evaluates how professional the dance appears. We ask 20 subjects to rate various aspects of the dance on a five-point scale. This qualitative analysis enables a comprehensive assessment of the dances, demonstrating that DanceEdit produces enhanced dance that is both professional and aligned with the users' intentions.

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